

Review of IP-CRM For Phase I Clinical Trials

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1 Introduction

Phase I clinical trials are the first stage of testing a new drug or treatment in humans. They primarily focus on evaluating the safety, dosage, and side effects of the treatment. The main goal of these trials is to determine the maximum tolerated dose (MTD) of the treatment which involves defining a target toxicity rate and determining the dosage with a toxicity probability closest to the target using a relatively few number of participants. Phase I trials often take several months and lay the foundation for further testing in Phase II and III trials, which focus more on efficacy and long-term safety. The article, “A Phase I Dose-Finding Design Incorporating Intra-Patient Dose Escalation,” by Beibei Guo and Suyu Liu proposes a novel method of conducting a phase I trial using a combination of intra-patient dose escalation and the Continual Reassessment Method (IP-CRM) [1].

1.1 Phase I Designs

The 3+3 method is the dose-escalation strategy most commonly used in Phase I clinical trials to determine the MTD of a new drug. The trial begins by administering a low dose to a small cohort of 3 patients, who are carefully monitored for any dose-limiting toxicities (DLTs) that could be severe enough to prevent further dose escalation. If no patient experiences DLTs, the dose is increased for the next cohort of 3 patients. However, if one patient in the group experiences a DLT, the trial proceeds with caution by adding 3 more patients at the same dose to better assess safety. If two or more patients experience DLTs, the dose level is considered too high, and the previous dose is established as the MTD. This method continues to escalate doses until the MTD is found, ensuring patient safety while trying to identify an effective therapeutic dose [2].

While the 3+3 design is widely used for its simplicity and straightforward approach, it does have limitations. One major drawback is that it can be inefficient in finding the optimal dose, as it may not always capture the true MTD due to the small number of patients at each dose level. The design can also be overly conservative, potentially leading to underdosing in some cases, as the escalation is based on DLTs in only a few patients [3]. Additionally, the 3+3 method does not provide detailed information on the overall dose-response relationship, which can make it challenging to identify the most effective dose for therapeutic benefit. As a result, newer dose-escalation methods, such as the Bayesian or model-based designs, are sometimes preferred for more precise and efficient determination of MTD [4].

The Continual Reassessment Method (CRM) is an alternative to the traditional 3+3 design. Unlike the 3+3 method, CRM is a model-based, adaptive dose-escalation approach that continuously updates the estimation of the MTD as data from patients are collected. The method begins by selecting an initial dose and enrolling a small cohort of patients. After observing the patients for any DLTs, the CRM model recalculates the probability of the MTD and adjusts the dose for the next cohort of patients accordingly. The goal is to find the dose

that has a pre-specified probability of causing DLTs, typically around 30%, based on real-time data from previous patients. The trial continues this process of adjusting doses and reassessing the MTD, allowing for a more flexible and precise determination of the appropriate dose compared to the more rigid 3+3 design [5].

The CRM offers several advantages, including more efficient dose escalation, better utilization of available patient data, and a higher likelihood of identifying the optimal therapeutic dose. It also avoids the inherent limitations of the 3+3 method, such as small cohort sizes and potential underdosing [6]. However, the method relies on statistical models that require careful design and calibration, and incorrect model assumptions or misestimations can lead to inaccurate dose determination. Additionally, CRM can be more complex to implement and interpret, requiring a robust data collection and analysis infrastructure. Despite these challenges, CRM is considered a more precise and adaptive approach for Phase I trials, particularly in cases where the 3+3 method may be too conservative or inefficient [7].

1.2 IP-CRM

The method proposed by Guo and Liu is the Intra-Patient Continuous Reassessment Method (IP-CRM). This is an adaptive design for Phase I clinical trials which incorporates patients receiving multiple dose levels over the course of the trial, allowing for intra-patient dose escalation based on their tolerance to toxicity. The IP-CRM design offers several advantages over traditional methods like 3+3. Since it allows for a greater amount of data points with few patients, it makes more efficient use of the available data and may identify the optimal dose more precisely. Additionally, by updating the MTD based on real-time data, the method adapts to the observed toxicity profile, potentially improving patient safety and trial efficiency [1]. However, the approach also comes with challenges, including the need for complex modeling and careful management of intra-patient correlations, which can make implementation more technically demanding. This review will cover the strengths and limitations of the proposed IP-CRM along with presenting a comparison of this method to more traditional Phase I designs through simulation.

2 IP-CRM Review

The core idea of the IP-CRM is to model the toxicity probabilities using a logistic regression model, where the toxicity probability at each dose level π_j is described by the equation:

$$\text{logit}(\pi_j) = \beta_0 + \beta_1 * \tilde{d}_j \quad (1)$$

Here, $\tilde{d}_j$ represents the effective dose for dose level j , and β_0 and β_1 are model parameters with $\beta_1 > 0$, assuming that toxicity increases with dose. The prior distributions for the model parameters should be determined using prior estimates of toxicity probabilities. At each interim stage of the trial, based on the data accumulated from previous patients, the MTD is re-estimated by calculating the posterior distribution of these parameters. The dose with the posterior mean toxicity probability closest to a pre-specified target toxicity rate is considered the MTD. Based on the logistic regression model, the likelihood function can be defined as:

$$(2) \quad L(\beta_0, \beta_1; D_n) = \prod_{j=1}^J \left[\prod_{i=1}^{n_j} \left(\pi_j^{y_{ij}} (1 - \pi_j)^{1-y_{ij}} \right) \right]$$

Where $D_n = \{(n_j, y_j), j = 1, \dots, J\}$ represents the data collected up to the n -th interim time, with n_j being the number of patients treated at dose level j , and y_{ij} being the indicator of toxicity for patient i at dose level j . π_j is the toxicity probability at dose level j , determined by the logistic model and J is the total number of dose levels under investigation. By Bayes' theorem, the joint posterior distribution of β_0 and β_1 is therefore:

$$(3) \quad f(\beta_0, \beta_1 | D_n) = f(\beta_0) f(\beta_1) L(\beta_0, \beta_1; D_n)$$

The posterior means of π_j at each dose level can be calculated and used to determine the MTD, the dose level at which the estimated toxicity probability is closest to the target toxicity rate. This is similar to the typical CRM except each patient in a cohort experiences intra-patient dose escalation until a toxicity is observed, which adds additional data points to the full toxicity data used to update the MTD estimate.

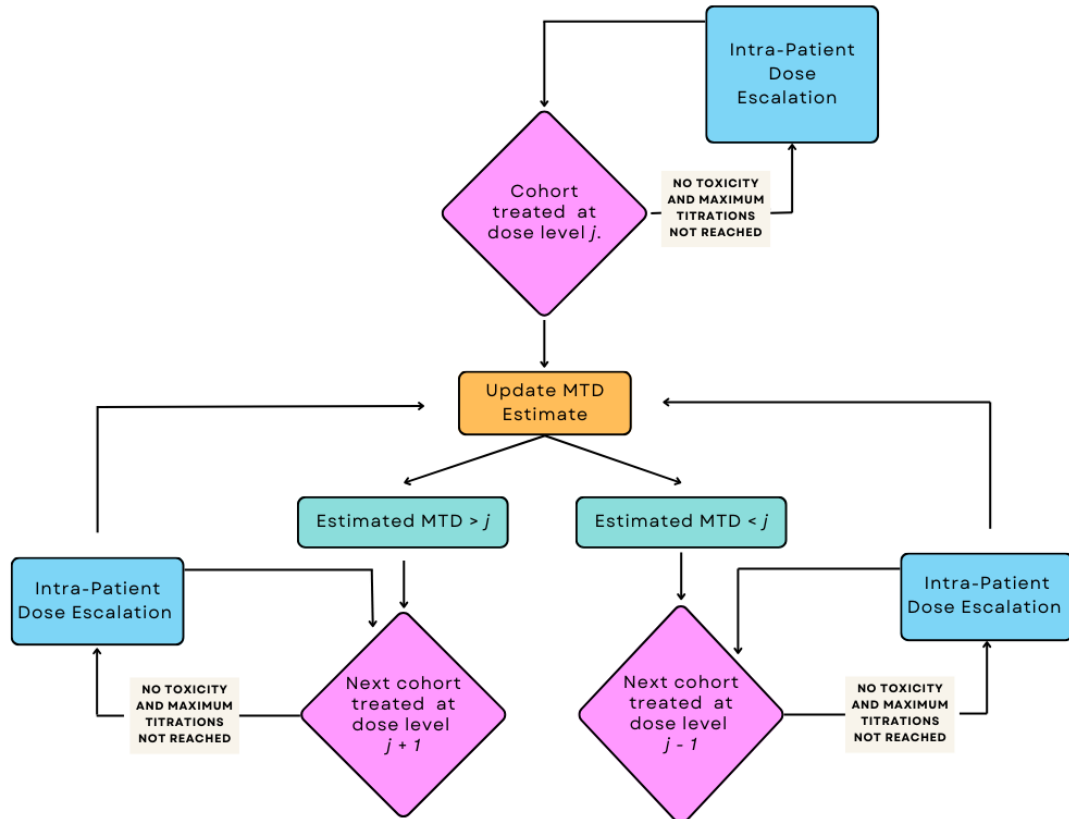


Fig. 1. Schematic showing the steps of the IP-CRM dose-finding strategy.

Figure 1 shows a schematic of the IP-CRM design. The first cohort of patients is treated at the lowest dose level in the first cycle. If a patient completes the cycle without experiencing toxicity, they will be escalated to the next higher dose in the subsequent cycle as

long as they have not reached the pre-specified maximum number of escalations. If a patient experiences toxicity, they will not receive any more treatment. When new patients are ready to be treated, the MTD is updated using all data from the previously treated patients. If the new MTD estimate exceeds the highest dose already tested, the new cohort will begin at the next dose level. If the updated MTD is lower than the highest dose already tested, the starting dose will be de-escalated for the next cohort. If the highest tested dose is equal to the MTD estimate, the starting dose will be the estimated MTD. If the lowest dose is determined to be too toxic based on a pre-specified threshold, the trial is terminated early to avoid further harm to patients. Once the maximum sample size is reached, the final MTD is determined.

It was illustrated in the article that the IP-CRM was able to reduce the sample size needed to determine the true MTD when compared to other Phase I designs in a simulation setting. The primary advantage of this method is that each patient can contribute multiple data points to enable estimation of MTD. The response adaptive design allows for more flexibility in dose escalation compared to the 3 + 3 method by adjusting doses based on previous patients' toxicity profiles and ensuring that doses are only escalated when deemed safe. The method prioritizes patient safety by minimizing exposure to harmful doses, carefully monitoring toxicity, and adjusting accordingly. The early termination rule further enhances patient safety by halting the trial if the lowest dose proves too toxic, preventing unnecessary risks. The use of posterior mean estimation for toxicity probability offers a more statistically informed and precise determination of the MTD, making it more reliable than traditional methods that often rely on arbitrary thresholds.

A major limitation of this design is that it does not account for intra-patient correlation. The authors propose adding a random effects term to the logistic model to incorporate patient-specific variability. In a real-world setting, it would be necessary to include this term because otherwise each patient's observations would be treated as independent, resulting in biased estimates. When sample size is small, a random effects model may not be well-identified and there may be difficulties in accurately estimating the variance components of the random effects. This can lead to unstable or unreliable conclusions about the MTD. Individual patients may not have similar responses to a treatment so for more generalizable results, data from multiple patients must be obtained even when conducting intra-patient dose escalation. This raises the question of whether the IP-CRM is truly able to produce generalizable MTD estimates with a very small sample size or if the estimates are largely overfitted.

Another consideration made by the authors is addressing potential carryover effects when conducting intra-patient dose escalation, which refers to a patient's response to an escalated dose being influenced by the previous doses administered to them. To combat this, it was proposed that a term $\beta_2 T_{ij}$ be added to the logistic model, T_{ij} with representing the cumulative dose patient i received prior to dose j . The carryover effect is represented by β_2 , which would be modeled based on a prior distribution. This adds complexity to the model specification as it would be difficult to accurately estimate the carryover effect without a sufficient amount of data to support the model. Estimating β_2 requires a good understanding of the underlying biological mechanisms that might cause such effects, which may not always be well understood or easily quantifiable.

Overall, the IP-CRM offers a more efficient and flexible approach to determining the MTD by reducing sample size and adjusting doses based on real-time data, which enhances patient

safety. However, the method's complexity increases when accounting for intra-patient correlations and potential carryover effects, which could lead to biased estimates if not properly modeled. Despite these challenges, the IP-CRM holds promise for more precise MTD estimation, though its generalizability in smaller sample sizes remains a key consideration.

3 Simulation Study

3.1 Simulation Methods

A simulation study was conducted to evaluate the performance of the IP-CRM in comparison to the standard 3+3 method and CRM in situations with a relatively small number of participants. The simulations were set up with the following parameters: 5 dose levels (J), a maximum of 9 patients (N), and a cohort size of 3 patients. The target toxicity rate was set to 0.3, representing the desired toxicity rate for identifying the MTD. Additionally, prior distributions were defined for the logistic regression parameters used in the Bayesian models of IP-CRM and CRM, with $\beta_0 \sim N(0,1)$ and $\beta_1 \sim \text{Exp}(1)$. A relatively uninformative distribution for the intercept term was chosen to avoid biasing the estimate for the dose effect, β_1 . The prior distribution for the dose effect was chosen based on what was specified in the original article [1]. A Gibbs sampling algorithm was used to sample from the posterior distribution of the model parameters using the 'R2jags' package in R. The maximum number of titrations for each patient was limited to 3, and an early termination toxicity threshold of 0.5 was applied for the IP-CRM method to stop the trial early if toxicity at the lowest dose level exceeded this threshold.

Four true toxicity scenarios were simulated, each representing different patterns of toxicity for the dose levels. In Scenario 1, the true MTD was at dose level 2 with increasing toxicity probabilities. Scenario 2 had the true MTD at dose level 3 and Scenario 3 had the MTD at dose level 4. Scenario 4 presented a non-monotonic increase in toxicity, with the true MTD at dose level 2 but with a less predictable pattern of toxicity across dose levels. Full descriptions of the true toxicity rates for each scenario are presented in Table 1. The simulation was conducted across 2000 trials for each of the four toxicity scenarios, and for each trial, the final MTD and safety index (the proportion of toxic patients) were recorded for each method.

Table 1. Simulated true toxicity rates.

Scenario	True toxicity rates respective to dose levels 1-5
1	0.1, 0.3 , 0.55, 0.65, 0.75
2	0.05, 0.15, 0.3 , 0.5, 0.8
3	0.02, 0.05, 0.1, 0.3 , 0.5
4	0.05, 0.3 , 0.2, 0.4, 0.6

3.2 Simulation Results

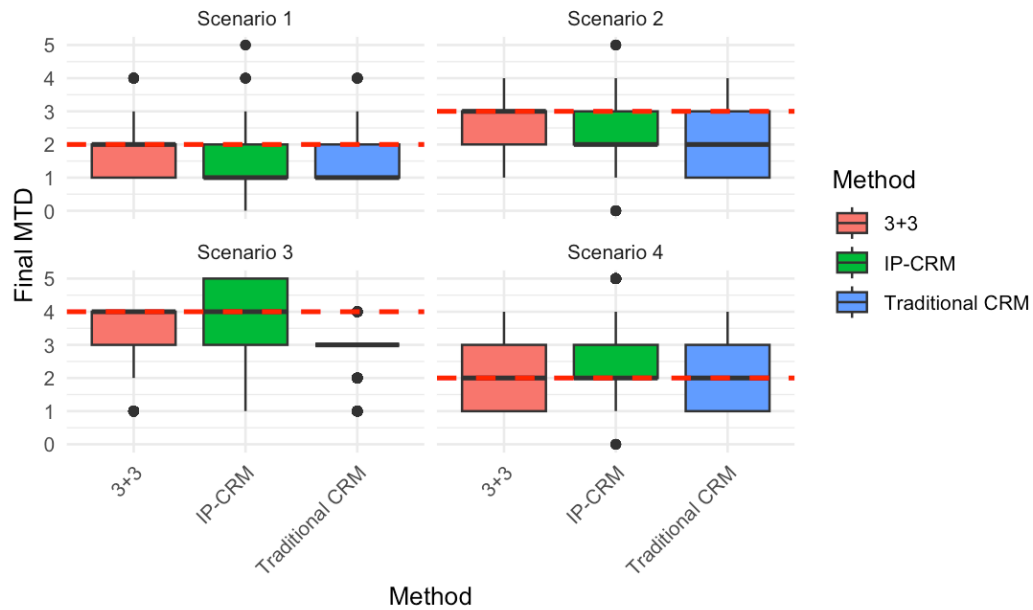


Fig. 2. Boxplot of final MTD estimates by method for each scenario. True MTD indicated by red dashed line.

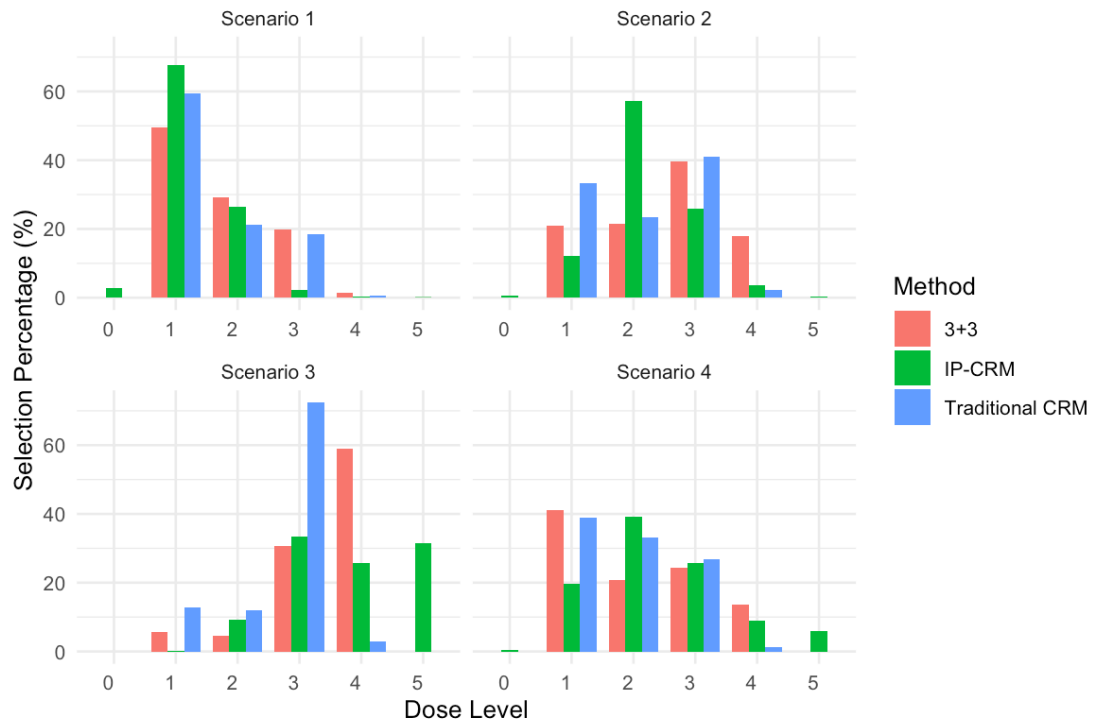


Fig. 3. Selection percentages of final MTD for each scenario by dose level and method. Dose level 0 represents cases where the trial was terminated early due to high toxicity at first dose.

Table 2. Mean safety indices for each method and scenario.

Method	Scenario 1	Scenario 2	Scenario 3	Scenario 4
3 + 3	0.269	0.167	0.064	0.218
CRM	0.132	0.077	0.028	0.104
IP-CRM	0.346	0.279	0.201	0.267

As shown by Figures 1 and 2, in Scenario 1, where the true MTD was at dose level 2, IP-CRM selected dose 1 most frequently (67.75%), followed by dose 2 (26.55%), with very few selections for higher doses. This indicates that IP-CRM tended to favor lower doses, even though dose 2 was the true MTD. CRM also selected dose 1 most often (59.55%) but also showed substantial selection of dose 2 (21.25%) and dose 3 (18.45%), indicating some variability and less precision. The 3+3 method selected dose 1 most frequently (49.50%) and dose 2 (29.20%), showing a greater degree of accuracy in selecting the true MTD, but it still over-selected dose 1 more than necessary.

In Scenario 2, where the true MTD was at dose level 3, IP-CRM selected dose 2 most frequently (57.30%), followed by dose 3 (25.90%), and dose 1 (12.25%). This indicates that IP-CRM underperformed and over-selected dose 2 compared to dose 3, the true MTD. CRM selected dose 3 most often (41.00%), but it also showed notable selection of doses 1 (33.35%) and 2 (23.30%), showing more accuracy and less precision. The 3+3 method selected dose 3 most frequently (39.60%), followed by doses 1 (20.85%) and 2 (21.60%), performing better than IP-CRM but with high variability.

In Scenario 3, where the true MTD was at dose level 4, IP-CRM selected dose 3 most frequently (33.40%), followed by dose 4 (25.65%), with dose 5 being selected 31.50% of the time. This suggests that IP-CRM performed well by selecting doses 3 and 4 near the true MTD, although it over-selected dose 5. CRM selected dose 3 most frequently (72.35%), making it less accurate in identifying the true MTD. The 3+3 method selected dose 4 most frequently (58.95%), followed by dose 3 (30.75%), showing the best accuracy.

In Scenario 4, where the true MTD was at dose level 2, IP-CRM selected dose 2 most frequently (39.10%), followed by dose 1 (19.65%) and dose 3 (25.85%). This suggests that IP-CRM performed reasonably well in identifying the true MTD in a case where the toxicity probability was non-monotonically increasing, which violated the model assumptions, but also showed some selection of lower doses. CRM selected dose 1 most often (38.80%), followed by dose 2 (33.10%), showing less accuracy in targeting the true MTD. The 3+3 method selected dose 1 most frequently (41.10%), with dose 2 selected 24.40% of the time, indicating less accuracy compared to IP-CRM.

The safety index, which represents the proportion of patients who exhibited toxicity, varied across the different methods and scenarios. The 3+3 method generally had a moderate safety index, with values ranging from 0.064 in Scenario 3 to 0.269 in Scenario 1. This indicated a higher toxicity rate in scenarios where lower doses were more frequently selected. The CRM method consistently had the lowest safety index, ranging from 0.028 in Scenario 3 to 0.132 in Scenario 1, suggesting that it resulted in fewer toxicities overall. The IP-CRM

method had the highest safety index across all scenarios, with values ranging from 0.201 in Scenario 3 to 0.346 in Scenario 1, indicating a higher proportion of patients experiencing toxicity due to its selection of higher doses.

4 Discussion

In this paper, we explored the performance of the IP-CRM for identifying the MTD in clinical trials through simulations with small sample size and a comparison with the CRM and the traditional 3+3 design. The IP-CRM method is a Bayesian approach that models toxicity probabilities using logistic regression, where dose escalation occurs within a patient rather than only between different cohorts. By incorporating intra-patient dose escalation, each patient can provide multiple data points, improving the statistical efficacy and reducing the overall sample size needed to identify the true MTD.

One of the primary strengths of the IP-CRM method is its ability to potentially reduce the sample size required to determine the MTD. Since each patient contributes multiple data points through intra-patient dose escalation, fewer patients are needed compared to traditional dose-finding designs, such as the 3+3 method. This is particularly advantageous in early-phase trials where sample size is often limited due to time and ethical constraints. The early termination rule, which halts the trial if the lowest dose is found to be too toxic, further improves patient safety. Furthermore, the use of posterior mean estimation for toxicity probability offers a more statistically informed and precise determination of the MTD, making it more reliable than traditional methods like 3+3 that rely on arbitrary thresholds.

Based on the simulation results, it is difficult to determine which method performed the best because each method had strengths and weaknesses in different scenarios. IP-CRM generally showed a good balance by selecting doses closer to the true MTD in most scenarios, but it tended to over-select lower doses, such as in Scenario 1 where it favored dose 1 even though dose 2 was the true MTD. CRM, on the other hand, showed more variability, with increased accuracy in certain scenarios but also substantial mis-selections, such as over-selecting dose 1 in Scenario 4, where the true MTD was at dose 2. The 3+3 method showed the least variability in some scenarios, like Scenario 3, where it selected dose 4 most frequently, but it still struggled with over-selection of lower doses, particularly in Scenarios 1 and 4.

Due to the nature of intra-patient dose escalation, the IP-CRM method had the highest proportion of observed toxicities as patients who observed no toxicity continued to have their dosage escalated, resulting in a higher probability of patients receiving higher doses in this method compared to the other two. IP-CRM, like other Bayesian methods, is sensitive to the prior assumptions made about the toxicity probability distribution. If those priors are mismatched to the actual data or are overly restrictive, it could lead to less flexibility in the dose-finding process, making it difficult for the model to adapt properly to the observed toxicity rates. It is likely in this case that the model was a poor fit due to the chosen priors for the parameters. In real-world scenarios, more complex sampling procedures would have to be employed to better estimate and refine the priors. These more complex sampling procedures can help the model adapt to the observed data more effectively and allow for greater flexibility in the dose-finding process.

The limitations of the IP-CRM method were highlighted in the simulation. One key issue is that the method does not account for intra-patient correlation, which could lead to biased estimates if patients' responses to treatment are not independent. This limitation is particularly concerning in real-world applications where individual patients may not have similar responses to treatment, and including a random effects term to model patient-specific variability could improve the method's accuracy. However, incorporating such terms could complicate the model and may be difficult to estimate with small sample sizes. Additionally, the method's sensitivity to prior assumptions about toxicity probabilities is a notable challenge, especially due to the small sample size. If the chosen priors are poorly matched to the actual data, the model may be too restrictive and unable to adapt properly to observed toxicity rates. This sensitivity makes the method less flexible in certain scenarios, which could lead to less accurate MTD estimates. In real-world settings, more complex sampling procedures might be required to better estimate and refine these priors, allowing the model to adapt more effectively. Finally, when implementing an IP-CRM design it is important to carefully select the maximum number of titrations a patient can receive in order to avoid excess toxicity. As shown in the simulation, IP-CRM can result in increased amounts of observed toxicities compared to other methods. The dosage at each level and other parameters should be specified after taking these issues into account.

Despite the results observed in this simulation study, Bayesian approaches, including methods like CRM, have been shown to consistently outperform the traditional 3+3 design in several cases [8]. In real-world scenarios, more research is needed to refine these types of models, particularly with respect to incorporating patient-specific variability and improving prior estimation techniques to ensure that the method provides reliable and generalizable results.

5 Conclusion

In conclusion, the IP-CRM method shows considerable promise in improving the effectiveness and safety of dose-finding in early-phase clinical trials by incorporating intra-patient dose escalation. This Bayesian approach, while potentially reducing sample size requirements and enhancing statistical precision, also faces some challenges, particularly with respect to prior assumptions, intra-patient correlation, and addressing potential carryover effects. The simulation results indicate that while IP-CRM offers a good balance in most scenarios, it can underperform or over-select low doses. Comparisons with the traditional 3+3 and CRM methods reveal that each approach has its strengths and weaknesses, and no method consistently outperformed the others across all scenarios.

The sensitivity of the IP-CRM model to prior distributions highlight areas for improvement. Future research should focus on refining the model, particularly by addressing these limitations, incorporating patient-specific variability, and exploring more advanced sampling techniques for better prior estimation. While Bayesian methods like IP-CRM offer substantial improvements over traditional dose-finding designs, further development is needed to enhance their reliability and generalizability in real-world clinical trial settings. Ultimately, the IP-CRM method holds significant potential for advancing the efficiency and safety of clinical trials, but continued refinement and research are essential for optimizing its application in diverse clinical scenarios.

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Appendix

```
library(tidyverse)
library(ggplot2)
library(R2jags)

# Define parameters
J <- 5 # Number of dose levels
N <- 9 # max patients
target_tox_rate <- 0.3
beta0_prior_mean <- 0
beta0_prior_sd <- 1
beta1_prior_rate <- 1
cohort_size <- 3
max_titrations <- 3
c <- 0.5 # Early termination threshold for high toxicity at the lowest dose

# Define initial skeleton of toxicity probabilities
skeleton <- c(0.02, 0.12, 0.3, 0.5, 0.65)

# True toxicity probs
scenario1 <- c(0.1, 0.3, 0.55, 0.65, 0.75) # true mtd at dose level 2
scenario2 <- c(0.05, 0.15, 0.3, 0.5, 0.8) # true mtd at dose level 3
scenario3 <- c(0.02, 0.05, 0.1, 0.3, 0.5) # true mtd at dose level 4
scenario4 <- c(0.05, 0.3, 0.2, 0.4, 0.6) # true mtd at dose level 2f but not monotonically increasing

simulate_data <- function(n_patients, dose_level, toxicity_probs) {
  tox_prob <- toxicity_probs[dose_level] # Probability of toxicity at this dose

  # Generate toxicities for each patient in the cohort
  toxicity <- rbinom(n_patients, 1, tox_prob) # Toxicity for each patient (1 = toxic, 0 = no toxic)

  # Create a dataframe with patient ID, dose level, and the simulated toxicity
  data <- data.frame(
    patient_id = 1:n_patients,
    dose_level = rep(dose_level, n_patients),
    toxicity = toxicity
  )

  return(data)
}

# Define the function to sample from the posterior distribution
```

```

gibbs <- function(data, beta0_prior_mean, beta0_prior_sd, beta1_prior_rate) {

  # Define the model in JAGS syntax
  model_code <- "
    model {
      # Prior distributions for beta0 and beta1
      beta0 ~ dnorm(beta0_prior_mean, 1 / (beta0_prior_sd^2))
      beta1 ~ dexp(beta1_prior_rate)

      # Likelihood
      for (i in 1:N) {
        toxicity[i] ~ dbern(p[i])
        logit(p[i]) <- beta0 + beta1 * dose_level[i]
      }
    }
  "

  # Define the data
  jags_data <- list(
    N = length(data$toxicity),
    toxicity = data$toxicity,
    dose_level = data$dose_level,
    beta0_prior_mean = beta0_prior_mean,
    beta0_prior_sd = beta0_prior_sd,
    beta1_prior_rate = beta1_prior_rate
  )

  # Define the parameters to monitor
  parameters <- c("beta0", "beta1")

  # Fit the model
  model_fit <- jags(data = jags_data,
    parameters.to.save = parameters,
    model.file = textConnection(model_code),
    n.chains = 4,
    n.iter = 1000,
    n.burnin = 200,
    n.thin = 2)

  beta0_samples <- model_fit$BUGSoutput$sims.list$beta0
  beta1_samples <- model_fit$BUGSoutput$sims.list$beta1

  # Return the posterior samples
  return(list(beta0_samples = beta0_samples, beta1_samples = beta1_samples))
}

```

```

# MTD Calculation

estimate_mtd <- function(beta0_samples, beta1_samples, target_tox_rate, J) {
  # Matrix to store the estimated toxicity probabilities for each dose level
  estimated_tox_probs <- matrix(0, nrow = length(beta0_samples), ncol = J)

  # Calculate estimated toxicity probabilities for each dose level using the posterior samples
  for (i in 1:J) {
    estimated_tox_probs[, i] <- 1 / (1 + exp(-(beta0_samples + beta1_samples * (i))))
  }

  # Calculate the mean toxicity probability for each dose level
  mean_tox_probs <- apply(estimated_tox_probs, 2, mean)
  print("Mean Toxicity Probabilities by Dose Level:")
  print(mean_tox_probs)

  # Find the dose level closest to the target toxicity rate
  closest_dose <- which.min(abs(mean_tox_probs - target_tox_rate))

  return(closest_dose)
}

#IP-CRM simulation
simulate_trial_ip_crm <- function(J, N, cohort_size, max_titrations, scenario,
                                target_tox_rate, c, skeleton) {
  # Initialize data frame to store results from all patients
  all_data <- data.frame(patient_id = integer(), dose_level = integer(),
                        toxicity = integer(), escalation_count = integer())
  dose_level <- 1 # Start with the lowest dose level
  cohort <- 1
  trial_terminated <- FALSE
  final_mtd <- 0
  toxic_patients <- 0 # Initialize the number of toxic patients

  while (cohort <= ceiling(N / cohort_size) && !trial_terminated) {
    print(paste("Cohort", cohort, "starting at dose level", dose_level))

    # Simulate the data for the current cohort
    data <- simulate_data(cohort_size, dose_level, scenario)

    # Count toxic patients for this cohort
    toxic_patients <- toxic_patients + sum(data$toxicity)

    # Initialize the escalation count for each patient (set to 0 initially)
    data$escalation_count <- 0

    # Add the data for the cohort to the overall trial data

```

```

all_data <- rbind(all_data, data)

# Check if the first dose level is too toxic (i.e., toxicity > c) for any patient
if (dose_level == 1 && mean(data$toxicity) > c) {
  print("Trial terminated early due to high toxicity at the first dose level.")
  trial_terminated <- TRUE
  break
}

# Implement intra-patient dose escalation: check toxicity and escalate doses for patients
for (i in 1:nrow(data)) {
  while (data$toxicity[i] == 0 && data$escalation_count[i] < max_titrations
        && data$dose_level[i] < J) {
    new_dose <- min(data$dose_level[i] + 1, J)
    print(paste("Patient", data$patient_id[i], "escalated from dose",
               data$dose_level[i], "to dose", new_dose))

    # Simulate toxicity at the new dose level
    data$toxicity[i] <- rbinom(1, 1, scenario[new_dose])

    # Update the patient's current dose and increment escalation count
    data$dose_level[i] <- new_dose
    data$escalation_count[i] <- data$escalation_count[i] + 1

    # Update all_data with the new treatment cycle results
    all_data <- rbind(all_data, data[i, ,drop = FALSE])

    # Count toxic patients for this escalation
    toxic_patients <- toxic_patients + sum(data$toxicity[i] == 1)
  }
}

# Re-estimate the MTD using logistic regression (based on all data so far)
gibbs_results <- gibbs(all_data, beta0_prior_mean, beta0_prior_sd,
                      beta1_prior_rate)
final_mtd <- estimate_mtd(gibbs_results$beta0_samples,
                         gibbs_results$beta1_samples, target_tox_rate, J)
print(paste("Updated MTD estimate: Dose level", final_mtd))

# Update the starting dose for the next cohort based on the new MTD
if (final_mtd > dose_level) {
  dose_level <- min(final_mtd, J)
} else if (final_mtd < dose_level) {
  dose_level <- max(final_mtd, 1)
} else {
  dose_level <- final_mtd
}

cohort <- cohort + 1
}

# Calculate the safety index: proportion of patients with toxic doses
safety_index <- toxic_patients / nrow(all_data)

```

```

return(list(all_data = all_data, final_mtd = final_mtd,
            safety_index = safety_index))
}

# Traditional CRM Simulation
simulate_trial_traditional_crm <- function(J, N, cohort_size, scenario,
                                           target_tox_rate, skeleton) {
  # Initialize data frame to store results from all patients
  all_data <- data.frame(patient_id = integer(), dose_level = integer(),
                        toxicity = integer())
  dose_level <- 1 # Start with the lowest dose level
  cohort <- 1
  trial_terminated <- FALSE
  toxic_patients <- 0 # Initialize the number of toxic patients

  while (cohort <= ceiling(N / cohort_size) && !trial_terminated) {
    print(paste("Cohort", cohort, "starting at dose level", dose_level))

    # Simulate the data for the current cohort
    data <- simulate_data(cohort_size, dose_level, scenario)

    # Count toxic patients for this cohort
    toxic_patients <- toxic_patients + sum(data$toxicity)

    # Add the data for the cohort to the overall trial data
    all_data <- rbind(all_data, data)

    # Re-estimate the MTD using logistic regression (based on all data so far)
    gibbs_results <- gibbs(all_data, beta0_prior_mean, beta0_prior_sd,
                          beta1_prior_rate)
    final_mtd <- estimate_mtd(gibbs_results$beta0_samples,
                             gibbs_results$beta1_samples, target_tox_rate, J)
    print(paste("Updated MTD estimate: Dose level", final_mtd))

    # Update the starting dose for the next cohort based on the new MTD
    if (final_mtd > dose_level) {
      dose_level <- min(final_mtd, J)
    } else if (final_mtd < dose_level) {
      dose_level <- max(final_mtd, 1)
    }

    cohort <- cohort + 1
  }

  # Calculate the safety index: proportion of patients with toxic doses
  safety_index <- toxic_patients / nrow(all_data)
  return(list(all_data = all_data, final_mtd = final_mtd,
            safety_index = safety_index))
}

```

```

# 3+3 method simulation
simulate_trial_3_plus_3 <- function(J, N, cohort_size, scenario, target_tox_rate) {
  # Initialize data frame to store results from all patients
  all_data <- data.frame(patient_id = integer(),
                        dose_level = integer(), toxicity = integer())
  dose_level <- 1 # Start with the lowest dose level
  cohort <- 1
  trial_terminated <- FALSE
  toxic_patients <- 0 # Initialize the number of toxic patients

  while (cohort <= ceiling(N / cohort_size) && !trial_terminated) {
    print(paste("Cohort", cohort, "starting at dose level", dose_level))

    # Simulate the data for the current cohort (3 patients at a time)
    data <- simulate_data(cohort_size, dose_level, scenario)

    # Count toxic patients for this cohort
    toxic_patients <- toxic_patients + sum(data$toxicity)

    # Add the data for the cohort to the overall trial data
    all_data <- rbind(all_data, data)

    # Apply 3+3 rule to determine the next dose level
    num_toxicities <- sum(data$toxicity)
    if (num_toxicities == 0) {
      dose_level <- min(dose_level + 1, J)
    } else if (num_toxicities == 1) {
      print("1 patient with toxicity, stay at the same dose level")
    } else {
      dose_level <- max(dose_level - 1, 1)
    }

    # If the dose level is 1 and too toxic, terminate the trial early
    if (dose_level == 1 && mean(data$toxicity) > target_tox_rate) {
      print("Trial terminated early due to high toxicity at the first dose level.")
      trial_terminated <- TRUE
    }

    cohort <- cohort + 1
  }

  # Calculate the safety index: proportion of patients with toxic doses
  safety_index <- toxic_patients / nrow(all_data)
  return(list(all_data = all_data, final_dose_level = dose_level,
             safety_index = safety_index))
}

```



```

# Initialize a list to store all results
all_results <- data.frame(
  method = character(),
  final_mtd = numeric(),
  safety_index = numeric(),
  scenario = character(),
  stringsAsFactors = FALSE
)

# Run the trials for each scenario
num_trials <- 2000
for (scenario_index in 1:4) {
  # Select the scenario based on the scenario index
  scenario <- list(scenario1, scenario2, scenario3, scenario4)[[scenario_index]]

  # Initialize lists to store results for each method
  ip_crm_results <- vector("list", num_trials)
  ip_crm_safety_indices <- numeric(num_trials)

  traditional_crm_results <- vector("list", num_trials)
  traditional_crm_safety_indices <- numeric(num_trials)

  results_3_plus_3 <- vector("list", num_trials)
  safety_indices_3_plus_3 <- numeric(num_trials)

  # Run IP-CRM simulation
  for (i in 1:num_trials) {
    ip_crm_results[[i]] <- simulate_trial_ip_crm(J, N, cohort_size, max_titrations,
                                                scenario, target_tox_rate, c, skeleton)
    ip_crm_safety_indices[i] <- ip_crm_results[[i]]$safety_index
  }

  # Run Traditional CRM simulation
  for (i in 1:num_trials) {
    traditional_crm_results[[i]] <- simulate_trial_traditional_crm(J, N,
                                                                    cohort_size,
                                                                    scenario,
                                                                    target_tox_rate,
                                                                    skeleton)
    traditional_crm_safety_indices[i] <- traditional_crm_results[[i]]$safety_index
  }

  # Run 3+3 simulation
  for (i in 1:num_trials) {
    results_3_plus_3[[i]] <- simulate_trial_3_plus_3(J, N, cohort_size, scenario,
                                                    target_tox_rate)
    safety_indices_3_plus_3[i] <- results_3_plus_3[[i]]$safety_index
  }

  # Extract final MTDs for each method
  ip_crm_mtds <- sapply(ip_crm_results, function(x) x$final_mtd)
  traditional_crm_mtds <- sapply(traditional_crm_results, function(x) x$final_mtd)
  mtds_3_plus_3 <- sapply(results_3_plus_3, function(x) x$final_dose_level)
}

```

```

# Combine results into a data frame for this scenario
scenario_df <- data.frame(
  method = rep(c("IP-CRM", "Traditional CRM", "3+3"), each = num_trials),
  safety_index = c(ip_crm_safety_indices, traditional_crm_safety_indices,
    safety_indices_3_plus_3),
  final_mtd = c(ip_crm_mtds, traditional_crm_mtds, mtds_3_plus_3),
  scenario = rep(paste("Scenario", scenario_index), num_trials * 3)
)

# Append the current scenario's results to the main results dataframe
all_results <- rbind(all_results, scenario_df)
}

#Visualization

true_mtd_values <- c(2, 3, 4, 2)

ggplot(all_results, aes(x = method, y = final_mtd, fill = method)) +
  geom_boxplot() +
  facet_wrap(~ scenario) + # Create separate plots for each scenario
  geom_hline(data = data.frame(scenario = paste("Scenario", 1:4),
    true_mtd = true_mtd_values),
    aes(yintercept = true_mtd), color = "red",
    linetype = "dashed", size = 1) +
  labs(
    title = "Final MTD Estimates by Method and Scenario",
    x = "Method",
    y = "Final MTD",
    fill = "Method"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Calculate the selection percentages for each dose level and method
selection_percentages <- all_results %>%
  group_by(scenario, method, final_mtd) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(scenario, method) %>%
  mutate(percentage = (count/2000) * 100) %>%
  ungroup() %>%
  # Complete all combinations of scenario, method, and final_mtd
  complete(scenario, method, final_mtd, fill = list(count = 0, percentage = 0)) %>%
  ungroup()

# Create the bar plot with consistent bar widths and show all dose levels
ggplot(selection_percentages, aes(x = factor(final_mtd), y = percentage, fill = method)) +
  geom_bar(stat = "identity", position = "dodge", width = 0.8) + # Set width for consistent bar size
  facet_wrap(~ scenario, scales = "free_x") +
  labs(
    title = "Selection Percentage by Dose Level and Method",

```

```

    x = "Dose Level",
    y = "Selection Percentage (%)",
    fill = "Method"
) +
theme_minimal() +
theme(axis.text.x = element_text(hjust = 1))

# Calculate the mean safety index for each scenario and method
safety_indices <- all_results %>%
  group_by(method, scenario) %>%
  summarise(safety_index = mean(safety_index))

```